

KENDRIYA VIDYALAYA SANGATHAN BHOPAL REGION

PRE-BOARD I SET 2

MARKING SCHEME SESSION 2024-25

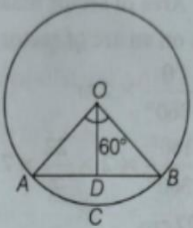
CLASS : - X SUBJECT :- MATHEMATICS(BASIC)

TIME :- 3 HOURS

MAX.MARKS: 80

Q.NO.	ANSWER	MARKS
1.	(a) 7 cm	1
2.	(b) 2500	1
3.	(b) $\frac{1}{2}$	1
4.	(d) coincident	1
5.	(c) 5	1
6.	(a) 12	1
7.	(d) 125^0	1
8.	(c) 13	1
9.	(b) $x(x - 5)$	1
10.	(d) 40^0	1
11.	(d) - 1	1
12.	(c) 7	1
13.	(a) 82	1
14.	(b) 462 cm^2	1
15.	(a) (0, - 3)	1
16.	(c) 45^0	1
17.	(b) 12 cm	1
18.	(d) IV quadrant	1
19.	(d) Assertion (A) is false but reason (R) is true.	1
20.	(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)	1
SECTION B		
21	$(10)^2 = (9 - x)^2 + (8 - 2)^2$ $100 = 81 - 18x + x^2 + 36$ $x^2 - 18x + 17 = 0$ $x^2 - 17x - x + 17 = 0$ $x(x - 17) - 1(x - 17) = 0$ $x = 17, x = 1$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

22	Let the ratio be $k : 1$ and point on X-axis is $(x, 0)$ $0 = \frac{k(-5) + 1(3)}{k+1}$ $-5k + 3 = 0$ $k = \frac{3}{5}$ so ratio is 3: 5 OR For perimeter of triangle find the lengths of sides $\text{Distance } AB^2 = (0 - 1)^2 + (0 - 0)^2$ $BC^2 = (1 - 0)^2 + (0 - 1)^2$ $AC^2 = (0 - 0)^2 + (1 - 0)^2$ So perimeter = $1 + 1 + \sqrt{2} = 2 + \sqrt{2}$	$\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
23	LCM of 48, 72 and 108 = $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$ $= 432$ Therefore, after 432 seconds, they will change simultaneously. We know that 60 sec. = 1 min. Therefore 432 sec. = $\frac{432}{60} = 7$ min. 12 sec. Hence, the lights change simultaneously at 7:07:12 am	1 $\frac{1}{2}$ $\frac{1}{2}$
24	$\frac{\sin 30^\circ + \tan 45^\circ}{\sec 30^\circ + \cot 45^\circ} = \frac{\frac{1}{2} + 1}{\frac{2}{\sqrt{3}} + 1}$ $= \frac{3}{2} \times \frac{\sqrt{3}}{2 + \sqrt{3}}$ $= \frac{3\sqrt{3}}{2(2 + \sqrt{3})} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = \frac{6\sqrt{3} - 9}{2(4 - 3)}$ $= \frac{3(2\sqrt{3} - 3)}{2}$	1 $\frac{1}{2}$ $\frac{1}{2}$
25	Let the angles be $a^\circ, (a + d)^\circ, (a + 2d)^\circ$ Then, we get $a + a + d + a + 2d = 180$ and $a + 2d = 2a$ after solving angles will be $40^\circ, 60^\circ, 80^\circ$ OR We know that 18 is the first multiple of 9 lies between 10 and 300 and last multiple of 9 up to 300 is 297. $a_n = a + (n - 1)d$ $297 = 18 + (n - 1)9$ $297 - 18 = (n - 1)9$ $n = 32$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
SECTION C		
26	Let $\sqrt{7}$ be a rational number. $\therefore \sqrt{7} = \frac{p}{q}$, where $q \neq 0$ and p & q are coprime. $7q^2 = p^2 \Rightarrow p^2 \text{ is divisible by } 7$ $\Rightarrow p \text{ is divisible by } 7 \text{ -----(i)}$	$\frac{1}{2}$ 1

	$\Rightarrow p=7a$, where 'a' is a positive integer $49a^2=7q^2 \Rightarrow q^2=7a^2 \Rightarrow q^2$ is divisible by 7 $\Rightarrow q$ is divisible by 7----(ii) (i) and (ii) lead to contradiction as 'p' and 'q' are coprime. $\therefore \sqrt{7}$ is an irrational number.	1 $\frac{1}{2}$
27	$\text{LHS} = \frac{\cos^2 A + (1 + \sin A)^2}{\cos A (1 + \sin A)}$ $\text{LHS} = \frac{\cos^2 A + \sin^2 A + 1 + 2 \sin A}{(1 + \sin A) \cos A}$ $\text{LHS} = \frac{1 + 1 + 2 \sin A}{(1 + \sin A) \cos A}$ $\text{LHS} = \frac{2(1 + \sin A)}{(1 + \sin A) \cos A}$ $\text{LHS} = \frac{2}{\cos A} = 2 \sec A$ OR STEP WISE MARKS FOR CORRECT SOLUTION	$\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
28	Chords AB subtends an angle 60° at the centre.  $\therefore \angle AOB = 60^\circ$ and $\angle OAB = \angle OBA = x$ [say] $[\because \text{angles opposite to equal sides are equal}]$ In $\triangle AOB$, $\angle AOB + \angle OBA + \angle OAB = 180^\circ$ $\Rightarrow 60^\circ + x + x = 180^\circ \Rightarrow x = 60^\circ$ So, $\triangle AOB$ is an equilateral triangle. Now, area of $\triangle AOB = \frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} \times (21)^2 = 190.95 \text{ cm}^2$ Area of sector OACBD = $\frac{\theta}{360^\circ} \times \pi r^2$ $= \frac{60^\circ}{360^\circ} \times 3.14 \times (21)^2 = 230.79 \text{ cm}^2$ $\therefore$ Area of minor segment ACBDA $= \text{Area of sector OACBO} - \text{Area of } \triangle AOB$ $= (230.79 - 190.95) \text{ cm}^2 = 39.84 \text{ cm}^2$	1 1 1
29	Given 2 and -3 are zeroes of $x^2 + (a+1)x + b$ On putting $x = 2$, we get $(2)^2 + (a+1)(2) + b = 0$ $4 + 2a + 2 + b = 0$ $2a + b = -6$(i) Again, on putting $x = -3$, we get $(-3)^2 + (a+1)(-3) + b = 0$ $9 - 3a - 3 + b = 0$ $-3a + b = -6$(ii) Solving (i) and (ii) we get $a = 0$ and $b = -6$ so $a + b = -6$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
30	Let one number be x.	

Then, another number = $(11 - x)$

According to question,

$$\frac{1}{x} + \frac{1}{11-x} = \frac{11}{28}$$

$$\frac{11-x+x}{x(11-x)} = \frac{11}{28}$$

$$\frac{11}{x(11-x)} = \frac{11}{28}$$

$$x(11-x) = 28$$

$$x^2 - 11x + 28 = 0$$

$$x = 4, x = 7$$

so two numbers are 4 and 7.

31

Since D, E, F are the mid points of BC, CA, AB respectively
Therefore, $EF \parallel BC$, $DF \parallel AC$, $DE \parallel AB$
BDEF is a parallelogram
 $\angle 1 = \angle 2$ & $\angle 3 = \angle 4$
 $\triangle FBD \sim \triangle DEF$
Also, DCEF is a parallelogram
 $\angle 3 = \angle 6$ & $\angle 1 = \angle 2$ (proved above)
 $\triangle DEF \sim \triangle ABC$

Since $PQ \parallel BC$ therefore $\triangle APR \sim \triangle ABD$
 $\Rightarrow \frac{AP}{AB} = \frac{PR}{BD}$ (i)

$\triangle AQR \sim \triangle ACD$
 $\Rightarrow \frac{AQ}{AC} = \frac{RQ}{DC}$ (ii)

Now, $\frac{AP}{AB} = \frac{AQ}{AC}$ (iii)

Using (i), (ii) & (iii), $\frac{PR}{BD} = \frac{RQ}{DC}$
But, $BD = DC$
 $\Rightarrow PR = RQ$ or AD bisects PQ

1

1

1

OR

1

1

1

SECTION D

32

Class (C.I.)	Frequency (f_i)	Class Mark (x_i)	$X_i f_i$	cf
40-45	8	42.5	340	8
45-50	9	47.5	427.5	17
50-55	10	52.5	525	27
55-60	9	57.5	517.5	36
60-65	5	62.5	312.5	41
65-70	4	67.5	270	45

$$\sum f_i = 45$$

$$\sum f_i x_i = 2392.5$$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{2392.5}{45} = 53.167$$

Size of $(N/2)$ th item is $45/2$ ie. 22.5

Which lies in the class interval 50-55

Here, $l = 50$, $h = 5$, $f = 10$, $cf = 17$

$$\text{Median} = l + \frac{\frac{N}{2} - cf}{f} \times h$$

$$= 50 + \frac{22.5 - 17}{10} \times 5 = 52.75$$

Therefore, Mean = 53.167 and Median is 52.75

OR

Monthly Expenditure	f_i	x_i	$f_i x_i$
1000-1500	24	1250	30000

Correct table
2marks

1

$\frac{1}{2}$

$\frac{1}{2}$

1

Correct table
2marks

1500-2000	40	1750	70000
2000-2500	33	2250	74250
2500-3000	x	2750	77000
3000-3500	30	3250	97500
3500-4000	22	3750	82500
4000-4500	16	4250	68000
4500-5000	7	4750	33250

$$172 + x = 200$$

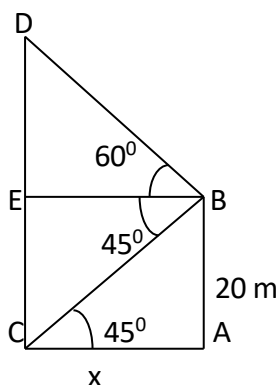
$$x = 28$$

$$\text{Mean} = \frac{532500}{200}$$

$$= 2662.5$$

1
½
1
½

33



Let AB be the building and CD be the tower

$$\text{Here } \tan 60^\circ = \sqrt{3} = \frac{h}{x}$$

$$h = x\sqrt{3} \dots\dots\dots(i)$$

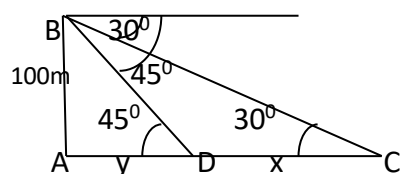
$$\tan 45^\circ = \frac{20}{x} = 1$$

$$x = 20 \text{ m} \dots\dots\dots(ii) \text{ (distance between tower and building)}$$

$$\text{solving (i) and (ii) to get } h = 20 \times 1.732 = 34.64 \text{ m}$$

$$\text{therefore, the height of the tower} = h + 20 = 54.64 \text{ m.}$$

OR



Let AB be the light house and C and D be the positions of ships.

$$\tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{100}{x+y}$$

$$x + y = 100\sqrt{3} \dots\dots\dots(i)$$

$$\tan 45^\circ = 1 = \frac{100}{y}$$

$$y = 100 \dots\dots\dots(ii)$$

$$\text{Solving (i) and (ii) to get } x = 100(\sqrt{3} - 1)$$

$$x = 100 \times .732$$

$$x = 73.2 \text{ m}$$

1 mark
for
correct
figure

1
½
1
½
½
½

1 mark
for
correct
figure

1
½
1
½

	distance between the ships is 73.2 m	1																		
34	Correct given, to prove, construction, figure Correct proof AR = AQ = 12 cm BP = BR = AB – AR = 3 cm CP = CQ = 7 cm BC = BP + PC = 3 + 7 = 10 cm	1 2 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$																		
35	Given, pair of linear equations is $3x - 4y + 3 = 0 \quad \dots(i)$ and $3x + 4y - 21 = 0 \quad \dots(ii)$ Table for $3x - 4y + 3 = 0$ or $y = \frac{3x+3}{4}$ is <table border="1"> <tr> <td>x</td><td>3</td><td>-1</td></tr> <tr> <td>$y = \frac{3x+3}{4}$</td><td>3</td><td>0</td></tr> <tr> <td>Points</td><td>A(3, 3)</td><td>B(-1, 0)</td></tr> </table> Plot the points A(3, 3) and B(-1, 0) on the graph paper and join them to get the straight line AB. Now, table for $3x + 4y - 21 = 0$ or $y = \frac{21-3x}{4}$ is <table border="1"> <tr> <td>x</td><td>3</td><td>7</td></tr> <tr> <td>$y = \frac{21-3x}{4}$</td><td>3</td><td>0</td></tr> <tr> <td>Points</td><td>A(3, 3)</td><td>C(7, 0)</td></tr> </table> Plot the points A(3, 3) and C(7, 0) on same graph paper and join them to get the straight line AC. The lines AB and AC intersect each other at point A(3, 3). So, the solution of the given pair of linear equations is (3, 3).	x	3	-1	$y = \frac{3x+3}{4}$	3	0	Points	A(3, 3)	B(-1, 0)	x	3	7	$y = \frac{21-3x}{4}$	3	0	Points	A(3, 3)	C(7, 0)	Correct points of eqn 1, 1 mark Correct points of eqn 2, 1 mark Correct plotting on graph 2 marks Correct solution 1 mark.
x	3	-1																		
$y = \frac{3x+3}{4}$	3	0																		
Points	A(3, 3)	B(-1, 0)																		
x	3	7																		
$y = \frac{21-3x}{4}$	3	0																		
Points	A(3, 3)	C(7, 0)																		
SECTION E																				
36.(i)	$l = \sqrt{r^2 + h^2}$ $l = \sqrt{6^2 + 8^2}$ $l = \sqrt{36 + 64}$ $l = \sqrt{100}$ $l = 10 \text{ m}$	$\frac{1}{2}$																		
(ii)	CSA of cone = $\pi r l$ $= \frac{22}{7} \times 6 \times 10$	$\frac{1}{2}$																		
(iii)(A)	$= 188.57 \text{ m}^2$	$\frac{1}{2}$																		
	CSA of cylinder = $2 \pi r h$ $= 2 \times \frac{22}{7} \times 6 \times 14$ $= 528 \text{ m}^2$	1																		
(iii)(B)	Cost of metal sheet used = 528×1000 $= ₹5,28,000$	1																		
	OR Volume of cylinder = $\pi r^2 h$ $= \frac{22}{7} \times (6)^2 \times 14$																			

	$\frac{AB}{DE} = \frac{BC}{EF} = \frac{\frac{BC}{2}}{\frac{EF}{2}} \frac{BM}{EN}$ Also $\angle B = \angle E$ Therefore $\triangle ABM \sim \triangle DEN$	1 $\frac{1}{2}$
	THE END	